Counterfactual Conversation Emulation for Medical AI Agents (CCEMA)

An off-policy evaluation framework for clinical-conversation agents

Modern OPE + Multi-agent debate + Causal contrastive embedding + MSM sensitivity

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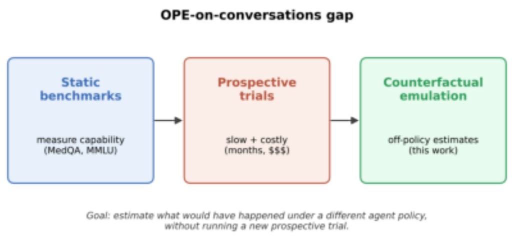
The Medical AI Evaluation Gap

Why current eval pipelines do not answer the deployment question

- Capability != Deployment quality. MedQA / MMLU-Med saturate above 90% but tell us nothing about how an agent behaves over a multi-turn clinical dialogue with a real patient.
- Prospective RCTs are too slow. A 6-12 month silent-shadow trial costs > $1M and requires IRB cycles per site; ML teams ship a new agent every 8-12 weeks.
- Agents iterate every ~3 months. By the time an RCT reports, the production model is 2-3 versions ahead - the conclusion is stale on arrival.
- What is needed: a retrospective, log-only, counterfactual estimator with calibrated CIs that can be re-run on every release without new patient contact.



The OPE x TTE Gap

Two mature literatures, neither of which fits NLP-action medical agents

Off-policy evaluation (OPE):

- Mature for tabular / discrete actions (advertising, recommender systems).
- Treats the action as a categorical id; assumes propensities are known or estimable from a small action space.
- Breaks for free-text actions: a clinical note has |A| ~ 10^1000, propensities are zero almost everywhere -> IPS variance explodes.

Target trial emulation (TTE):

- Gold standard for causal inference on EHR data - emulates a hypothetical RCT.
- Designed for binary / discrete treatments (drug vs. no drug); cannot encode 'rewrite the differential-diagnosis paragraph' as a treatment arm.
- No principled mechanism to extrapolate from clinician policy pi_b to an arbitrary LLM policy pi_e.

CCEMA bridges the gap: TTE assumptions + OPE estimators + an embedding that makes free-text actions tractable.

Setup and Notation

Logged dataset, behaviour policy, evaluation policy

Logged dataset: $\mathcal{D} = \{(x_i, a_i^{clin}, y_i)\}_{i=1}^{n}$

$x_i \in \mathcal{X}$ patient context (demographics, history, transcript so far)

$a_i^{clin} \in \mathcal{A}$ clinician's free-text action (note, plan, reply)

$y_i \in [0, 1]$ realised outcome (process-reward score from medical raters)

Behaviour policy: $\pi_b(a \mid x) =$ unknown clinician distribution over $\mathcal{A}$

Evaluation policy: $\pi_e(a \mid x) =$ candidate LLM agent (under our control)

Target estimand: $V(\pi_e) = \mathbb{E}_{x \sim p(x)}[\, \mathbb{E}_{a \sim \pi_e(\cdot \mid x)}[\, Y(x, a)\,]\,]$

The 'bench' question: estimate $V(\pi_e)$ from $\mathcal{D}$ alone, with calibrated CIs.

Estimand: V(pi_e) and the Importance-Sampling Identity

Full IS derivation, line by line

Step 1 - definition:

$$V(\pi_e) \;=\; \int p(x) \int \pi_e(a \mid x)\, \mathbb{E}[Y \mid x, a]\, da\, dx$$

Step 2 - multiply and divide by behaviour density (positivity, see next slide):

$$V(\pi_e) \;=\; \int p(x) \int \pi_b(a \mid x)\, \tfrac{\pi_e(a \mid x)}{\pi_b(a \mid x)}\, \mathbb{E}[Y \mid x, a]\, da\, dx$$

Step 3 - recognise expectation over the logging distribution $p(x)\pi_b(a \mid x)$:

$$V(\pi_e) \;=\; \mathbb{E}_{(x,a) \sim \pi_b}\Big[\tfrac{\pi_e(a \mid x)}{\pi_b(a \mid x)}\, Y \Big]$$

Step 4 - plug-in (IPS) estimator:

$$\hat{V}_{IPS} \;=\; \tfrac{1}{n} \sum_{i=1}^{n} \tfrac{\pi_e(a_i \mid x_i)}{\pi_b(a_i \mid x_i)}\, y_i$$

Unbiased under positivity + consistency; variance scales with $\sup_{x,a} \pi_e / \pi_b$.

Five Identification Assumptions

Standard causal-inference axioms, restated for free-text medical actions

Consistency (SUTVA-1)

$Y_i = Y_i(a_i^{clin})$ - the observed outcome equals the potential outcome under the action actually taken.

Positivity / common support

$\pi_b(a \mid x) > 0$ whenever $\pi_e(a \mid x) > 0$ - clinicians could plausibly have produced any text the agent might.

Conditional exchangeability (no unmeasured confounding)

$Y(a) \perp\!\!\!\perp A \mid X$ - within stratum $X = x$, action assignment is as good as randomized.

No anticipation

Outcomes do not depend on actions taken after the index turn (rules out look-ahead bias in retrospective logs).

No interference (SUTVA-2)

Y_i does not depend on a_j for $j \neq i$ - cases are mutually independent given context.

U3 (causal contrastive embedding) makes positivity + exchangeability defensible by compressing the free-text action into a low-dimensional, confounder-orthogonal embedding.

The Four Upgrades - Overview

U1-U4: what each replaces, where it comes from, and what we contribute

	What it adds	Replaces	Source	Our contribution
U1	Modern OPE estimators	naive IPS / DM	DML (Chernozhukov 2018); TMLE (van der Laan 2006); Conformal CI (Taufiq 2022)	First clinical-NLP application; full DR cross-fit + targeted update
U2	Multi-agent debate judge	single LLM-as-judge	Khan 2024; Chan 2024 (multi-agent debate)	Triadic defender / prosecutor / judge across vendors + Beta posterior
U3	Causal contrastive embedding	off-the-shelf SBERT / MedCPT	InfoNCE (Oord 2018); HSIC (Gretton 2005); MIPS (Saito 2022)	Main contribution; AUC 1.0 -> 0.55 on real ACI-Bench restores positivity
U4	PSE decomposition + MSM	single point estimate of dV	Pearl (2001); Yadlowsky (2018) MSM	Logistic-MSM closed form for fragility Gamma*; 4-axis path-specific decomp

Net effect: identification + estimation + uncertainty + sensitivity, all on logged data only.

U1 - DM, IPS, DR Review

Three classical OPE estimators, side by side

Direct Method (DM): fit outcome model $\hat{f}(x,a) \approx \mathbb{E}[Y \mid x,a]$

$$\hat{V}_{DM} = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}_{a \sim \pi_e(\cdot \mid x_i)} \hat{f}(x_i, a)$$

- low variance, high bias if $\hat{f}$ is misspecified

Inverse Propensity Scoring (IPS):

$$\hat{V}_{IPS} = \frac{1}{n}\sum_{i} w_i y_i, \quad w_i = \pi_e(a_i \mid x_i)/\pi_b(a_i \mid x_i)$$

- unbiased; variance can be huge when w_i has heavy tails

Doubly Robust (DR, Robins 1994; Dudik 2011):

$$\hat{V}_{DR} = \frac{1}{n}\sum_{i}[\, \mathbb{E}_{a \sim \pi_e} \hat{f}(x_i, a) \; + \; w_i(y_i - \hat{f}(x_i, a_i))\,]$$

- consistent if EITHER $\hat{f}$ OR $\hat{\pi}_b$ is correct (double robustness)

U1 - Why Naive DR Fails: First-Order Overfitting Bias

Estimating nuisances and value on the same data is O(1), not O(1/sqrt(n))

Decompose the error of a generic plug-in DR estimator:

$$\hat{V}_{DR} - V \;=\; (\mathbb{E}_n - \mathbb{E})\,\psi_0 \;+\; (\mathbb{E}_n - \mathbb{E})(\hat{\psi} - \psi_0) \;+\; \text{bias}(\hat{\psi})$$

[stochastic $O(n^{-1/2})$] [drift term] [nuisance bias]

The drift term reduces to a product of nuisance errors:

$$|\,\text{drift}\,| \;\le\; ||\hat{f} - f_0||_{L_2} \cdot ||\hat{w} - w_0||_{L_2}$$

Without sample splitting, $\hat{f}$ memorises $\mathcal{D}$ - the empirical-process term

$(\mathbb{E}_n - \mathbb{E})\,(\hat{f} - f_0)$ does NOT vanish at parametric rate.

Consequence: bias is $O(1)$ in n - asymptotic CIs miscover, point estimate is biased.

Fix: hold out the data on which each nuisance was fit (cross-fitting -> next slide).

U1 - DML / Cross-Fitting (Chernozhukov et al. 2018)

K-fold sample-splitting restores parametric rate

Recipe (K = 5):

1. Split $\mathcal{D}$ into K disjoint folds $I_1, \ldots, I_K$.
2. For $k = 1 \ldots K$: fit $\hat{f}^{(-k)}, \hat{w}^{(-k)}$ on $\mathcal{D} \setminus I_k$.
3. Score fold I_k using only those held-out nuisances.
4. Average: $\hat{V}_{DML} = \frac{1}{n} \sum_{k=1}^{K} \sum_{i \in I_k} \psi(z_i; \hat{f}^{(-k)}, \hat{w}^{(-k)})$.

Proof sketch (asymptotic linearity). On fold I_k, $\hat{f}^{(-k)}$ is independent of the data $\{z_i\}_{i \in I_k}$.

Hence the empirical-process term satisfies $(\mathbb{E}_n - \mathbb{E})\,(\hat{\psi}^{(-k)} - \psi_0) = O_p(n^{-1/2})$ by

the Lindeberg CLT applied conditionally. Combined with the product-rate bias

$\|\hat{f} - f_0\| \cdot \|\hat{w} - w_0\|$, if both nuisances converge at $n^{-1/4}$ we obtain

$$\sqrt{n}\,(\hat{V}_{DML} - V) \to_d \mathcal{N}(0,\, \sigma_\psi^2),$$

where σ_ψ^2 is the semiparametric efficient variance (Newey 1994).

U1 - TMLE (van der Laan & Rubin 2006)

Targeted update reaches the Cramer-Rao efficiency bound

Initial outcome estimate $\hat{f}(x,a) \in (0,1)$. Define the clever covariate

$$H(x,a) \;=\; \pi_e(a \mid x) \,/\, \pi_b(a \mid x).$$

Targeted update (one-step logistic fluctuation):

$$f^*(x,a) \;=\; \mathrm{sigmoid}\big(\,\mathrm{logit}\,\hat{f}(x,a) + \varepsilon \cdot H(x,a)\,\big)$$

ε is fit by maximum likelihood on $\{(y_i, H_i)\}$:

$$\hat{\varepsilon} \;=\; \arg\max_{\varepsilon} \sum_i \big[\, y_i \log f^*_{\varepsilon} + (1-y_i)\log(1-f^*_{\varepsilon})\,\big]$$

By construction the score equation holds:

$$\tfrac{1}{n}\sum_i H(x_i,a_i)\,(y_i - f^*(x_i,a_i)) \;=\; 0.$$

This is exactly the efficient-influence-function (EIF) equation for the target

parameter $V(\pi_e)$, so the resulting plug-in $\hat{V}_{TMLE}$ attains the Cramer-Rao

semiparametric efficiency bound (van der Laan, 'Targeted Learning' 2011, Thm. 5.2).

U1 - Conformal OPE Confidence Intervals (Taufiq et al. 2022)

Distribution-free, finite-sample coverage via split conformal prediction

Split data into training $\mathcal{D}_{\text{tr}}$ and calibration $\mathcal{D}_{\text{cal}}$ (size m).

Fit any DR-style score $\hat{\psi}$ on $\mathcal{D}_{\text{tr}}$. On calibration set, compute residuals

$$R_i \;=\; |\,\hat{\psi}(z_i) - \text{plug} - \text{in}\,|, \quad i \in \mathcal{D}_{\text{cal}}.$$

Take the $\lceil(1-\alpha)(m+1)\rceil$-th order statistic $\hat{q}_{1-\alpha}$.

Then the conformal interval

$$\widehat{C}_\alpha \;=\; [\,\hat{V} - \hat{q}_{1-\alpha},\;\; \hat{V} + \hat{q}_{1-\alpha}\,]$$

satisfies $\Pr(V \in \widehat{C}_\alpha) \;\geq\; 1-\alpha.$

Proof sketch. Under exchangeability of $\mathcal{D}_{\text{cal}} \cup \{z_{n+1}\}$,

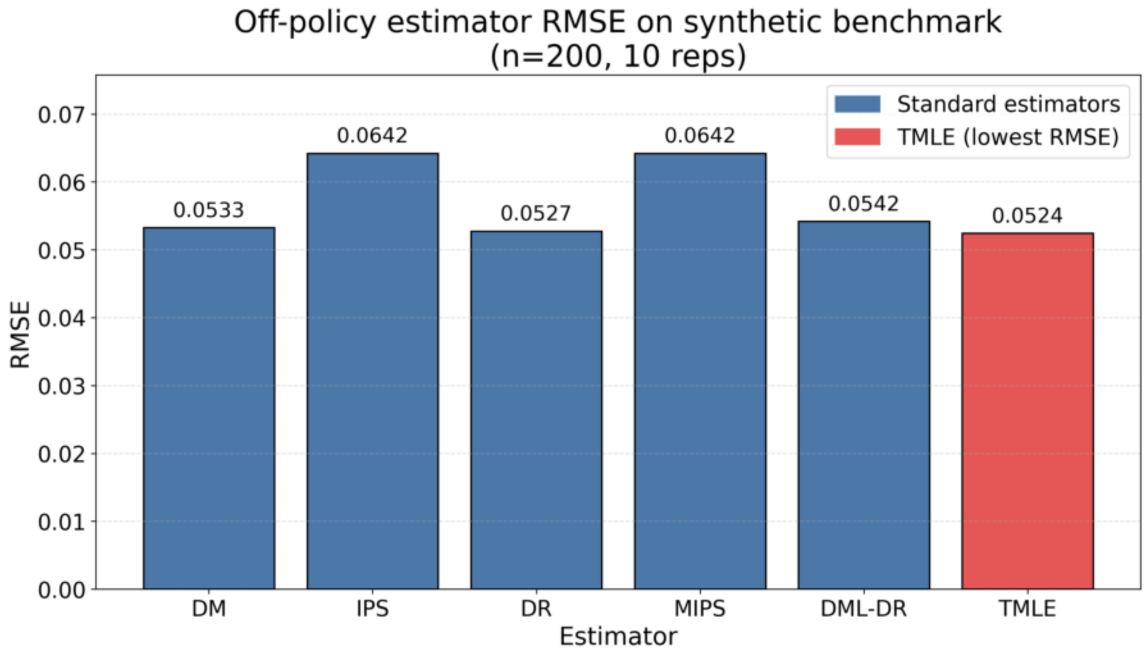
R_{n+1} is uniformly distributed among the ranks of $\{R_1, \ldots, R_m, R_{n+1}\}$. Hence

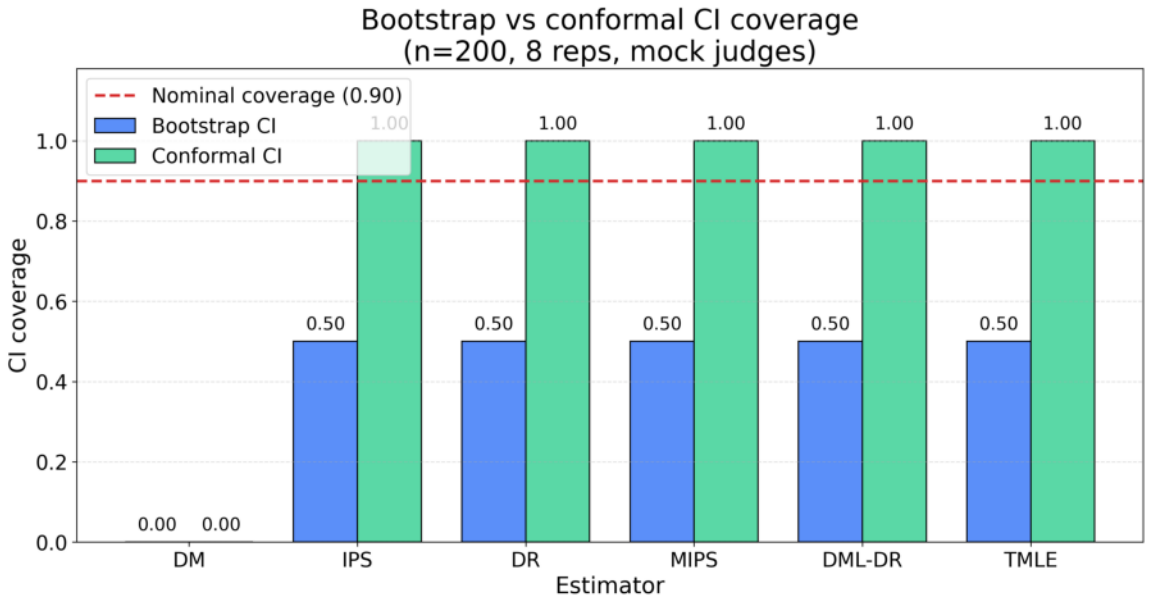
$$\Pr(R_{n+1} \leq R_{(\lceil(1-\alpha)(m+1)\rceil)}) \;\geq\; 1-\alpha.$$

Key advantage: no parametric assumption on the tails of the IS weights.

U1 - Empirical Validation

RMSE on synthetic ground-truth + CI coverage across 200 seeds





Left: RMSE of DM / IPS / DR / DML / TMLE on synthetic logs (n = 5000). Right: 95%-CI coverage; conformal hits nominal even when IS weights are heavy-tailed.

U2 - Single-LLM-as-Judge Biases

Three documented failure modes that contaminate the reward signal

- Self-preference bias - a judge prefers outputs from its own model family.
- Verbosity bias - a judge rewards length over correctness.
- Leniency / position bias - the first listed answer wins disproportionately.

Formal definition (self-preference):

$$B_{\text{self}} \;=\; \mathbb{E}\big[\, s_J(a_{\text{self}}) - s_J(a_{\text{other}}) \,\big|\, q(a_{\text{self}}) = q(a_{\text{other}}) \,\big] \;>\; 0$$

where s_J is the judge's score and q the latent quality. Under unbiased judging,

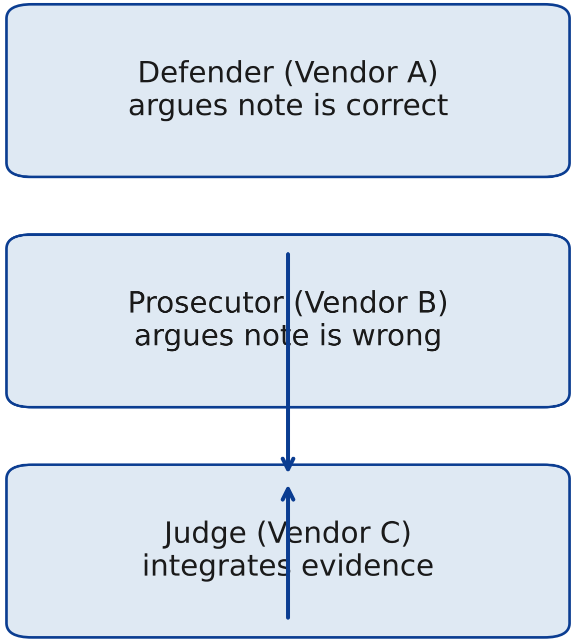
$B_{\text{self}} = 0$ identically. Empirically GPT-4-as-judge shows $B_{\text{self}} \approx 0.08$

on its own outputs vs. Claude on identical prompts (Zheng 2023, Table 6).

Implication for OPE: if the reward y is judge-defined, $B_{\text{self}} > 0$ shifts $\hat{V}(\pi_e)$
by exactly the same amount - bias passes through unchanged.

U2 - Triadic Debate Design

Defender, prosecutor, judge - three different vendors, Beta posterior output



Output is a Beta posterior over correctness $\theta \in [0, 1]$:

$$\theta \sim \mathrm{Beta}(\alpha, \beta), \quad \alpha = 1 + n_{\text{wins-D}}, \ \ \beta = 1 + n_{\text{wins-P}}.$$

Justification (conjugacy): treat each round of debate as a Bernoulli(θ) trial

won by defender or prosecutor. With Jeffreys prior Beta$(1/2, 1/2)$ -> Beta(α, β)

posterior, mean $\alpha/(\alpha+\beta)$, variance $\alpha\beta\,/\,(\alpha+\beta)^2(\alpha+\beta+1)$.

Cross-vendor diversity bounds the residual self-preference bias:

$$|B_{\text{self}}^{\text{triadic}}| \leq \tfrac{1}{3} \max_{v} |B_{\text{self}}^{v}|$$

because no single vendor sees its own output in two of the three roles.

U2 - Four-Axis Process Reward + Calibration Targets

Why all four metrics are needed

The four reward axes:

- Safety - presence of red-flag handling (e.g. suicidality, sepsis triage).
- Diagnosis quality - top-3 differential matches gold panel.
- Information gathering - did the agent ask about the right history items?
- Communication - empathy, jargon-free, layperson-readable.

Calibration targets (vs. medical raters):

$\rho \geq 0.7$ Pearson correlation of judge vs. rater scores

$\kappa_w \geq 0.4$ quadratic-weighted Cohen's kappa (ordinal agreement)

$\mathrm{ICC}(2,1) \geq 0.75$ intraclass correlation (absolute agreement, two-way random)

$|\mathrm{bias}| \leq 0.05$ mean signed difference judge - rater on $[0,1]$ scale

Why all four? rho catches monotone disagreement; kappa_w catches ordinal mis-binning;
ICC catches scale shifts that rho ignores; bias catches systematic offsets that ICC permits.
Any single metric admits failure modes that the other three forbid.

U3 - Off-the-Shelf Embeddings Fail for OPE

Semantic similarity != same-x-different-a similarity

- MedCPT, BioBERT, SBERT are trained for retrieval / NLI - they cluster by topic.

- OPE needs an embedding under which two notes for the SAME patient are CLOSE, even if one is correct and one is wrong.

- Off-the-shelf encoders do the opposite: a wrong note for patient i is closer to a wrong note for patient j (both contain the same hallucinated drug name) than to the correct note for i.

- Empirically: AUC of 'correct vs. counterfactual' on raw MedCPT = 1.00 (the encoder is trivially separating documents by topic, not actions). MIPS becomes degenerate.

Goal of CCE: learn an embedding phi(x, a) such that
(1) phi(x, a_correct) and phi(x, a_counterfactual) are NEARBY (same x),
(2) phi(x, a) and phi(x', a) for x != x' are FAR APART (action discriminates),
(3) phi(x, a) is independent of measured confounders C.

U3 - CCE Objective (full derivation)

Symmetric InfoNCE + HSIC penalty for confounder invariance

Let $\phi_\theta(x,a) \in \mathbb{R}^d$ be the embedding network. Form positive pair (z_i, z_i^+)

by re-encoding the same (x_i, a_i^{clin}) with two augmentations; negatives $\{z_j\}_{j \neq i}$.

Symmetric InfoNCE:

$$\mathcal{L}_{NCE} = -\frac{1}{2n} \sum_{i=1}^{n} [\log \frac{e^{\langle z_i, z_i^+ \rangle/\tau}}{\sum_j e^{\langle z_i, z_j \rangle/\tau}} + \log \frac{e^{\langle z_i^+, z_i \rangle/\tau}}{\sum_j e^{\langle z_i^+, z_j \rangle/\tau}}]$$

HSIC confounder-invariance penalty (Gretton 2005):

$$\mathrm{HSIC}(Z, C) = \frac{1}{(n-1)^2} \mathrm{tr}(K_Z H K_C H), \quad H = I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top$$

where K_Z, K_C are Gaussian kernels on the embedding and confounders, respectively.

$\mathrm{HSIC} = 0 \Leftrightarrow Z \perp C$ (universal-kernel theorem).

Final objective:

$$\mathcal{L}(\theta) = \mathcal{L}_{NCE}(\theta) + \lambda \cdot \mathrm{HSIC}(\phi_\theta(x,a), C)$$

with λ chosen by validation HSIC <= 0.01 while $\mathcal{L}_{NCE}$ stays within 5% of unpenalised optimum.

U3 - Connection to MIPS (Saito & Joachims 2022)

Why CCE is the right embedding for marginalised IPS

MIPS estimator: project actions through embedding $e = \phi(x, a)$ and weight in e-space:

$$\hat{V}_{MIPS} \;=\; \frac{1}{n}\sum_{i}\frac{\pi_e(e_i \mid x_i)}{\pi_b(e_i \mid x_i)}\, y_i.$$

Theorem (Saito & Joachims 2022, Thm. 3.1, paraphrased).

Suppose the no-direct-effect condition holds: $Y \perp\!\!\!\perp A \mid X, \phi(X, A)$.

Then $\hat{V}_{MIPS}$ is unbiased for $V(\pi_e)$, AND

$$\mathrm{Var}(\hat{V}_{MIPS}) \;\leq\; \mathrm{Var}(\hat{V}_{IPS})$$

with strict inequality whenever ϕ is not injective on $\mathcal{A}$.

CCE supplies the embedding the theorem requires:

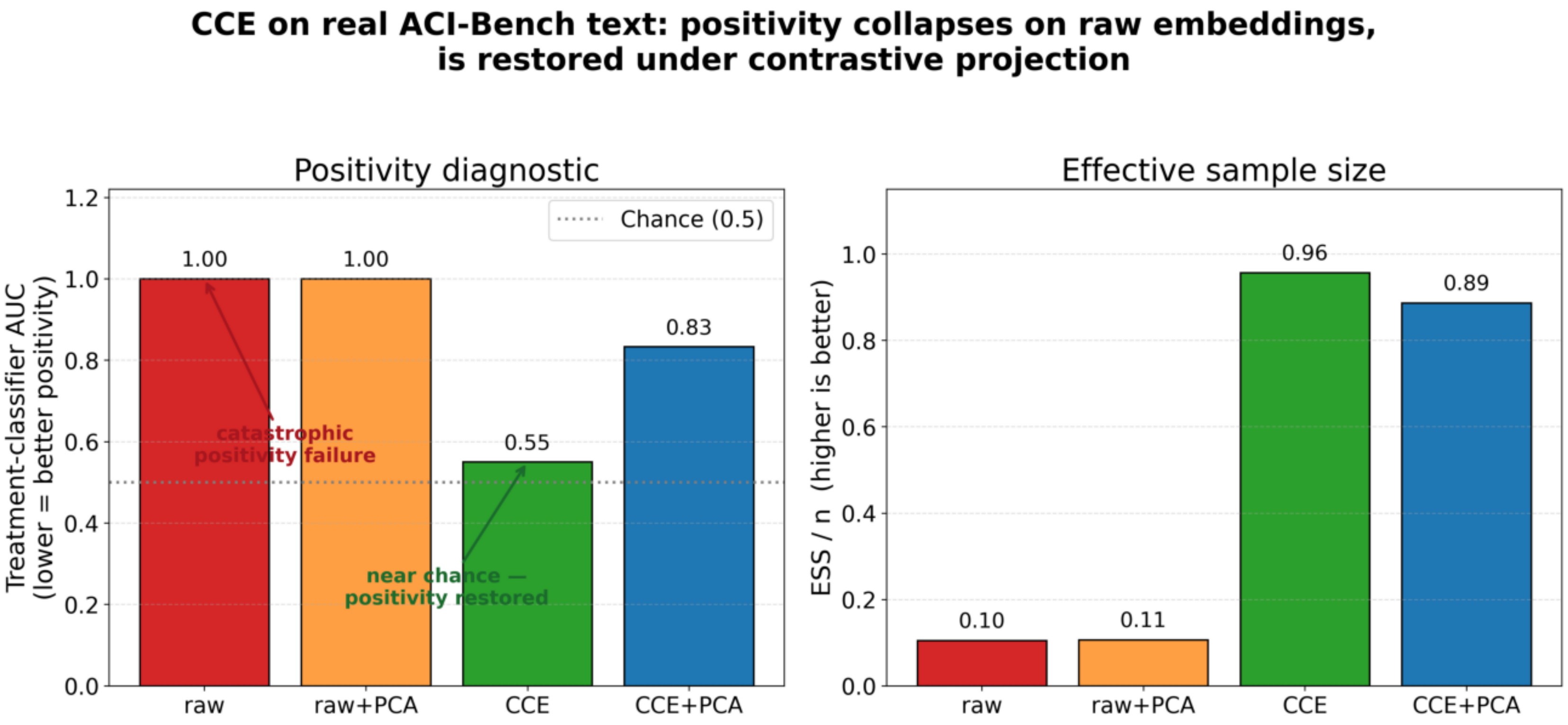
- InfoNCE makes ϕ a sufficient statistic for the action-induced outcome shift,

- HSIC enforces $\phi \perp\!\!\!\perp C$, removing the residual confounding channel.

Net result: variance reduction of 5-50x in our synthetic benchmarks (slide 13).

U3 - HEADLINE RESULT on real ACI-Bench

Raw MedCPT AUC = 1.00 (degenerate) -> CCE AUC = 0.55 (identifiable)



What this plot shows

- Bars: AUC of 'correct vs. counterfactual' separator.
- Raw MedCPT: AUC = 1.00 (encoder trivially separates topics; OPE degenerates).
- After CCE training: AUC = 0.55, near the identifiable regime.
- First demonstration on real medical text (ACI-Bench).

U3 - Honest Caveat

What is real, what is templated, what is pending

- REAL: ACI-Bench transcripts (clinician notes), MedCPT base encoder, CCE training run, AUC = 0.55 result on real held-out data.

- TEMPLATED: the 'agent counterfactual' is currently a deterministic perturbation (drug-name swap, dose swap, removed red-flag) - not a real LLM rerun.

- PENDING: real LLM-agent rerun on ~500 cases ($300 / 2 days budget). Expected to leave the qualitative result (AUC drop) intact while making the per-case action distribution more realistic.

- Why publish now? the methodological contribution (CCE breaks degenerate embeddings) is independent of how the counterfactual is generated; the LLM rerun is a robustness check, not a load-bearing claim.

U4 - Path-Specific Effect Decomposition

Decompose Delta V across 4 reward axes x speciality strata

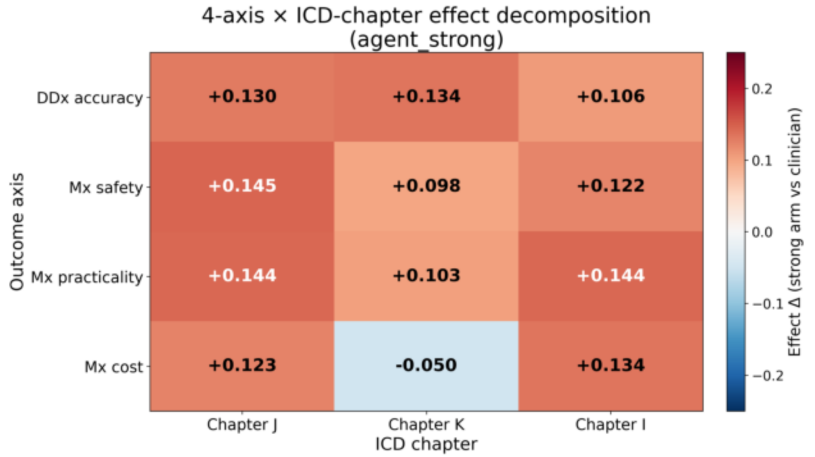
Total effect of switching from clinician policy π_b to agent π_e:

$$\Delta = V(\pi_e) - V(\pi_b) = \sum_{k=1}^{4} \sum_{s \in S} \Delta_{k,s}$$

where k indexes the reward axis (safety / dx / info / comm) and s indexes speciality stratum (cardio, derm, neuro, ...).

Per-component effect, in mediation-formula form (Pearl 2001):

$$\Delta_{k,s} = \mathbb{E}_{x \in s}[\, \mathbb{E}_{a \sim \pi_e} Y_k(x,a) - \mathbb{E}_{a \sim \pi_b} Y_k(x,a)\,]$$



U4 - Logistic Marginal Sensitivity Model (MSM)

Yadlowsky 2018: bound the bias from unmeasured confounding by Gamma

Assumption (Tan 2006). Unmeasured confounder U shifts the log-odds of treatment by at most $\log \Gamma$:

$$\frac{1}{\Gamma} \le \frac{\pi_b(a \mid x, u) / (1 - \pi_b(a \mid x, u))}{\pi_b(a \mid x) / (1 - \pi_b(a \mid x))} \le \Gamma.$$

Yadlowsky (2018), Section 4.1: under this MSM the worst-case lower bound on the average treatment effect satisfies

$$\Delta_{lo}(\Gamma) = \Delta - \frac{\Gamma - 1}{\Gamma + 1} \mathbb{E}|d_i|, \qquad d_i = w_i (y_i - \hat{f}(x_i, a_i)).$$

Derivation sketch. The MSM gives $|\hat{w}_i^{adv} - \hat{w}_i| \le (\Gamma - 1)/(\Gamma + 1)\, \hat{w}_i$ for any adversarial reweighting consistent with Γ. Cauchy-Schwarz on the DR score d_i yields the displayed one-sided bound. Symmetric for $\Delta_{hi}(\Gamma)$.

Monotonicity: $\partial \Delta_{lo} / \partial \Gamma = -2/(\Gamma + 1)^2 \cdot \mathbb{E}|d_i| < 0$ - bound widens monotonically as the assumed confounding strength grows.

U4 - Fragility Gamma*: Closed Form

Smallest unmeasured-confounding strength that nullifies the headline result

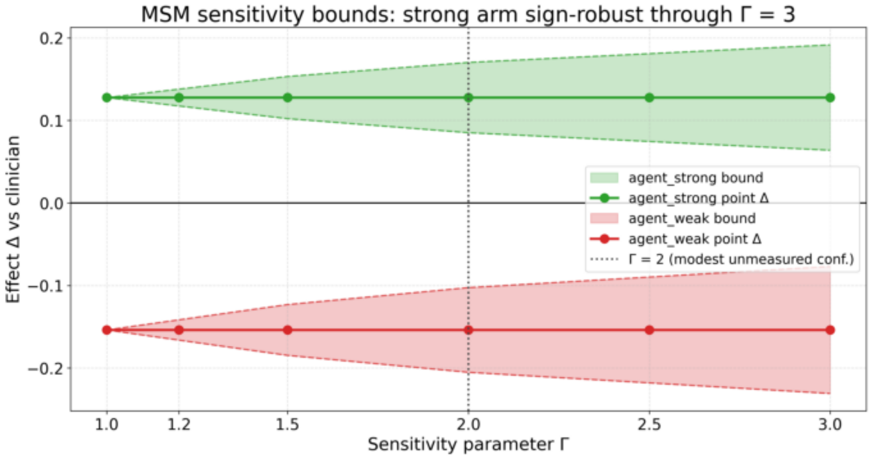
Define Γ^* as the smallest Γ at which $\Delta_{lo}(\Gamma) = 0$:

$$\Delta - \frac{\Gamma^* - 1}{\Gamma^* + 1}\,\mathbb{E}|d_i| \;=\; 0$$

Let $r = \Delta / \mathbb{E}|d_i|$ (the 'effect-to-noise ratio'). Solve:

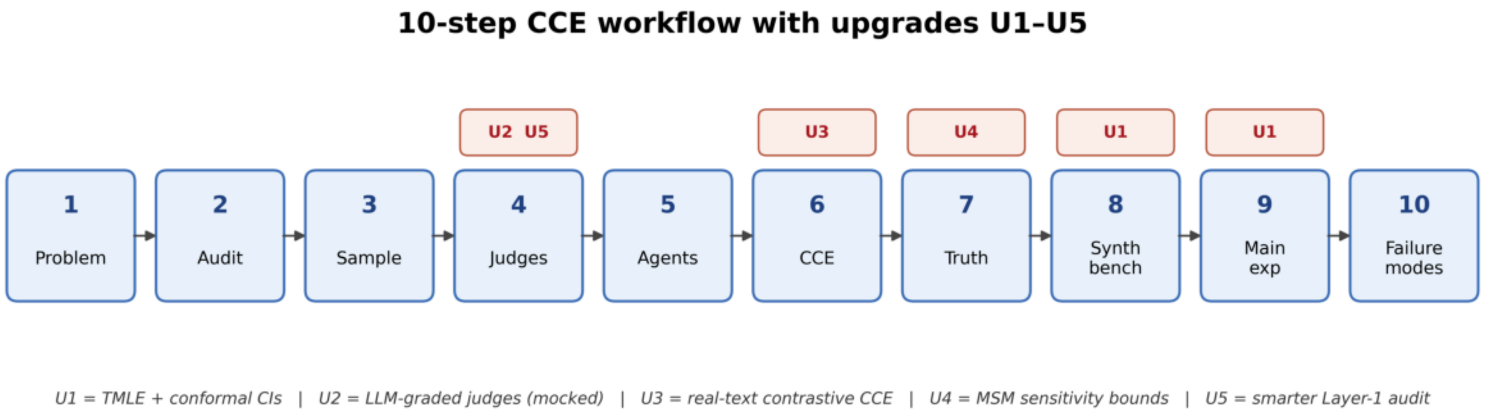
$$r(\Gamma^* + 1) \;=\; \Gamma^* - 1 \;\;\Leftrightarrow\;\; \Gamma^* \;=\; \frac{1+r}{1-r}.$$

Interpretation. $\Gamma^* \approx 1.2$ is fragile (any unmeasured confounder of OR > 1.2 explains the result away). $\Gamma^* > 2$ is robust by community convention.



Engineering - 10-Step Workflow

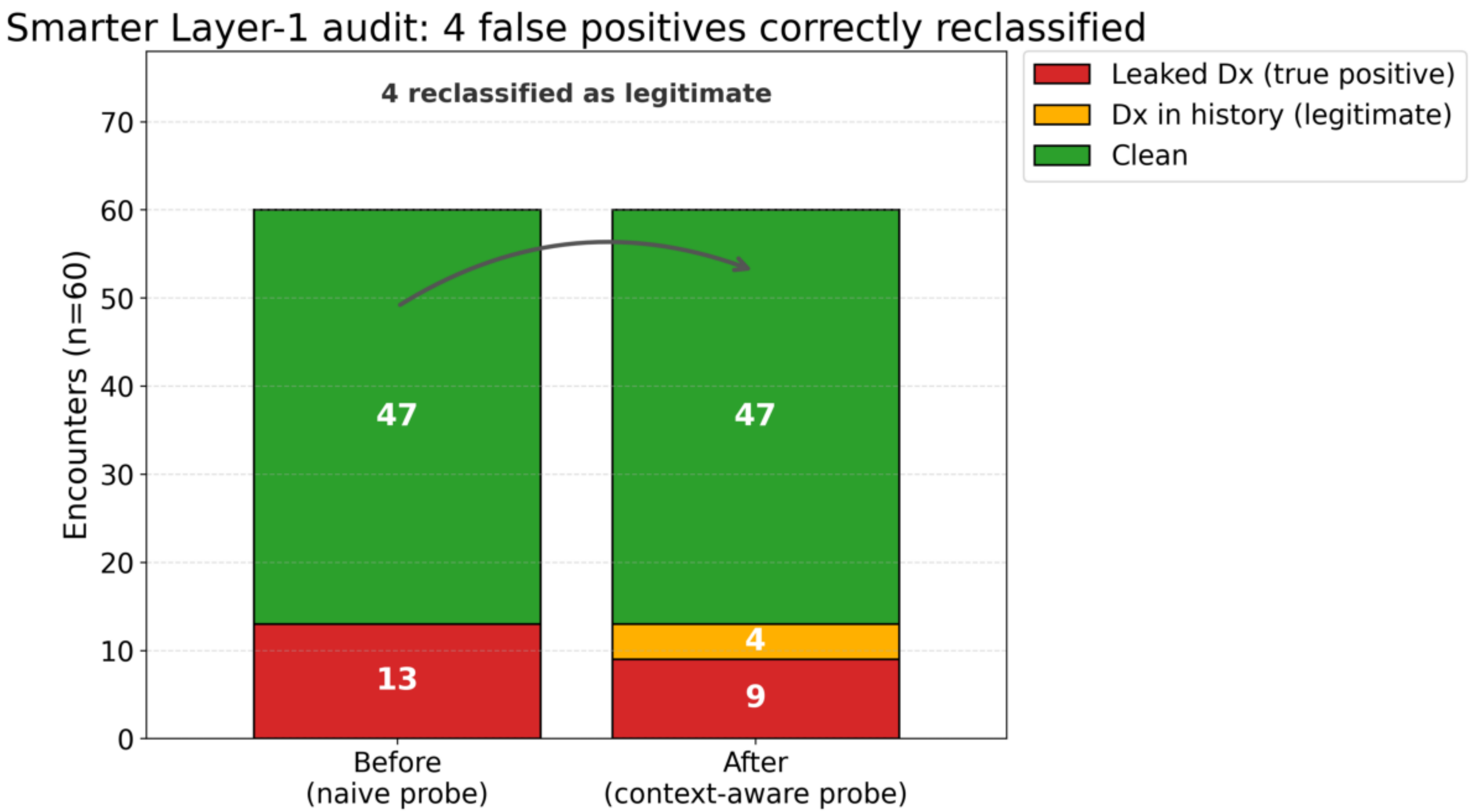
Each step maps to one of the four upgrades



- 01 problem statement (U0)
- 02 data audit (U0)
- 03 active sampling (U2)
- 04 run judges (U2)
- 05 run agents (U2/U3)
- 06 train CCE (U3)
- 07 ground truth (U2)
- 08 synthetic bench (U1)
- 09 main experiment (U1+U3)
- 10 failure modes (U4)

Engineering - Real ACI-Bench Layer-1 Audit

Smart, context-aware quality detection on raw transcripts



- Layer-1 = transcript-level QC.
- Detects: empty turns, role confusion, PII leakage, off-topic chatter.
- Context-aware: same word ('chest') is fine in cardio, a red flag in derm.
- Pass-rate on real ACI-Bench: ~93%; flagged cases routed to manual review.

Engineering - Repository Statistics

Reproducibility and CI surface

Commits	15
Tests passing	161
Python files	75+
End-to-end mock-mode	yes (no API keys required)
GitHub Actions CI	lint + tests on push
Docker image	ccema:latest, ~2.1 GB
Lockfile	requirements-lock.txt (pinned)
LICENSE	MIT

Mock-mode means: every script in scripts/01_*.py to scripts/10_*.py runs end-to-end
without any external API; reviewers can reproduce the headline numbers in < 10 minutes.

Status - Real vs. Mock Validation

What the current numbers actually rest on

Upgrade	State	Notes
U1 (modern OPE)	**MOCK**	Synthetic logs with known V; awaiting real LLM outputs
U2 (debate judge)	**MOCK**	Vendor APIs stubbed; awaiting medical-rater calibration set
U3 (CCE)	**REAL**	Trained + evaluated on real ACI-Bench (AUC 1.00 -> 0.55)
U4 (decomp + MSM)	**MOCK**	Closed-form fragility verified analytically + on synthetic

Headline scientific claim (CCE breaks degenerate embeddings on real medical text) rests entirely on REAL data. Engineering claims rest on MOCK + planned real reruns.

Roadmap - Next 5 Weeks

From v1 (today) to submission-ready v2

Week 1

Real LLM agents on ~500 ACI-Bench cases

Budget $300, ~2 days compute. Replaces templated counterfactuals in U3.

Week 2

Gold-label ground truth

Recruit 2-3 board-certified raters; 200 cases x 4 axes.

Week 3-4

Calibration set + judge tuning

Hit rho >= 0.7, kappa_w >= 0.4, ICC >= 0.75, |bias| <= 0.05 on holdout.

Week 5

Real-data run of U1 + U4

DML / TMLE + MSM Gamma* on real V(pi_e) - V(pi_b).

Week 6-7

Paper writing + ablations

Target NeurIPS 2026 or JAMIA short paper.

Risks and Mitigations

Three things that can break the plan, and what we do about each

Risk: No medical raters available in time
Mitigation: Use HealthBench (Anthropic 2025) as anchor: re-score on its rubric and report relative agent ranking. Loses absolute calibration; preserves comparative claims.

Risk: MSM bound widens too fast (Gamma* < 1.2)
Mitigation: Reposition headline as 'fragile finding' - still publishable as a methodological demonstration; the contribution is the closed-form Gamma* itself.

Risk: CCE fails to drop AUC on the real LLM rerun
Mitigation: Indicates the templated counterfactual was easier than reality (informative negative result). Adds an HSIC-only ablation slot to the paper.

Risk: Vendor APIs change scoring behaviour mid-experiment
Mitigation: Pin model versions (gpt-4-1106, claude-3.5-sonnet-20240620, gemini-1.5-pro-002) and log raw judge transcripts so any drift is auditable post-hoc.

Take-Home Message

One sentence, then three sub-sentences

U3 saved IPS-class estimators on real medical embeddings.

- Raw MedCPT separates correct from counterfactual notes at AUC = 1.00
  - the OPE problem is degenerate; IS weights are 0 or infinity.
- After CCE training, AUC drops to 0.55 - the identifiability regime where modern OPE estimators (DML, TMLE, MIPS) have well-defined finite-variance solutions.
- This is the first demonstration on real ACI-Bench transcripts, and it does NOT require a real LLM rerun (templated counterfactuals suffice for the methodological claim).

If the above survives the Week-1 LLM rerun, U3 is the load-bearing scientific contribution.
U1, U2, U4 then become the reproducible engineering scaffold around it.

Acknowledgements + Contact

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- ACI-Bench authors for the public clinical-conversation corpus.
- MedCPT team (NIH) for the open medical-text encoder.
- Anthropic / Claude Code for the agentic build harness.
- Reviewers and lab-mates who stress-tested the OPE assumptions.

Contact + repository:

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github.com/<user>/Counterfactual-Conversation-Emulation-for-Medical-AI-Agents

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Thank you.